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Analog Astronaut Training Experiences: A Statistical Approach to Astronaut Readiness

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Abstract

Astronaut training is a complex, multi-faceted process integrating technical expertise, psychological resilience, and adaptability to extreme environments. With the increasing complexity of human spaceflight missions, ensuring optimal astronaut preparedness is critical to mission success. Analog missions, which simulate space conditions in terrestrial environments, have gained recognition as essential preparatory experiences. However, their direct impact on astronaut readiness and mission performance remains an area requiring further quantitative assessment.

This study employs a statistical approach to evaluate the role of analog training in astronaut readiness, leveraging an expanded database of astronaut profiles. By analyzing mission histories, training backgrounds, and self-reported preparedness levels, we assess the correlation between analog mission participation and overall spaceflight performance. Additionally, we explore key operational competencies such as teamwork, problem-solving, and decision-making under simulated mission constraints. The results provide valuable insights into the significance of analog training in enhancing astronaut capabilities and offer recommendations for optimizing future astronaut training programs. These findings contribute to the ongoing discourse on astronaut selection and preparation, reinforcing the importance of terrestrial simulations in shaping the next generation of space explorers.

Keywords: Analog, Readiness, Performance, Simulation, Training, Statistics

Acronyms/Abbreviations

NASA - National Aeronautics and Space Administration
ESA - European Space Agency
ISS - International Space Station
LEO - Low Earth Orbit
ICE - Isolated, Confined, and Extreme (environments)
NEEMO - NASA Extreme Environment Mission Operations
HI-SEAS - Hawai'i Space Exploration Analog and Simulation
EVA - Extravehicular Activity
CHAPEA - Crew Health and Performance Exploration Analog
CAVES - Cooperative Adventure for Valuing and Exercising human behaviour and performance Skills (ESA training)
THOR - Tracing Habitability, Organics, and Resources
HSA - Hybrid Space Analog
OKRs - Objectives and Key Results
SMART - Specific, Measurable, Achievable, Relevant, Time-bound
PDCA - Plan-Do-Check-Act
NASA-TLX - NASA Task Load Index
PANAS - Positive and Negative Affect Schedule
VR - Virtual Reality
AR - Augmented Reality
AI - Artificial Intelligence
AMI - Analog Mission Impact
SGAC - Space Generation Advisory Council
DIVINAS - Diversity in Astronaut Selection

1. Introduction

Human spaceflight is undergoing a significant transformation in the 21st century. The rise of commercial launch providers, renewed governmental investment in exploration, and increasing participation from emerging space nations have reshaped the astronautical landscape [1]. Programs such as NASA's Artemis [2], ESA's Moon initiatives, and multinational Mars exploration roadmaps highlight the shift toward sustained lunar presence and eventual interplanetary missions. These frontiers demand new approaches to astronaut preparation, as future missions will involve longer durations, greater isolation, and increasingly multicultural crews [3].

Historically, astronaut training evolved in tandem with mission demands. Mercury and Apollo emphasized survival and engineering proficiency. The Shuttle era expanded toward multidisciplinary expertise. The International Space Station (ISS) fostered global collaboration and international crew operations [4]. Today, the complexity of exploration missions

highlights the need to redefine astronaut readiness for future decades.

1.1 Background and Motivation

While traditional training programs remain highly effective, analog missions (Earth-based simulations that mimic aspects of spaceflight) have gained recognition as complementary training environments [5]. Programs such as NEEMO, HI-SEAS, and Concordia have demonstrated their ability to replicate isolation, confinement, and operational stressors [6][7]. Despite strong qualitative support, systematic and quantitative evaluation of their contribution to astronaut readiness remains limited [8].

1. Defining Astronaut Readiness in the 21st Century

The construct of "astronaut readiness" has undergone a paradigm shift, evolving from an early emphasis on superlative physical fitness and technical skill to a holistic, multi-dimensional requirement for the 21st century. This evolution is not merely operational but is profoundly shaped by societal changes. As detailed by Williams, Santos, et al. (2024), the archetype of the astronaut has radically transformed since 1961, moving from a narrow selection of military test pilots to a diverse corps of scientists, engineers, and specialists valued for their adaptability, intercultural competence, and psychological resilience.

This shift, driven by the goals of long-duration, international, and commercial exploration beyond Low Earth Orbit (LEO), demands a new definition of readiness. It now represents the synergistic manifestation of competencies required to operate effectively in the uniquely hostile and isolated environment of space, accurately described as Isolated, Confined, and Extreme (ICE) (Suedfeld & Steel, 2000; Pagel & Choukèr, 2016). This section deconstructs this modern construct into its four core, interdependent pillars: Technical, Psychological, Interpersonal, and Physical, and argues that achieving integrated readiness across all domains necessitates a strategic and synergistic use of terrestrial, orbital, and planetary analog missions.

2.1 The Four Pillars of Integrated Readiness

The historical analysis by Williams et al. (2024) provides empirical evidence for the emergence of these four pillars, demonstrating how selection criteria have explicitly broadened to encompass them. Modern astronaut readiness can therefore be conceptualized through these interdependent domains, as summarized in Table 1. Mastery in one domain is insufficient

without competence in the others, as mission success depends on their integration.

Table 1: Pillars of 21st-Century Astronaut Readiness

Pillar	Competencies	Challenges	Literature
Technical & Operational	Systems mastery, procedural execution, emergency response, scientific experimentation, maintenance & repair, autonomous decision-making.	Increased autonomy due to communication delays, novel systems, lack of rapid abort options.	Sandal, 2001
Psychological Resilience	Stress tolerance, emotional regulation, adaptability, concentration, motivation, coping with monotony and danger.	Extreme isolation, confinement, "Earth-out-of-view" phenomenon, constant risk.	Suedfeld & Steel, 2000; Kanas & Manzey, 2008; Pagel & Choukèr, 2016
Interpersonal & Team	Communication, conflict resolution, leadership/followership, cultural competence, trust, group decision-making.	Multinational crews, prolonged confinement, autonomy from mission control.	Kanas & Manzey, 2008; Sandal et al., 2011
Physical Resilience	Microgravity/partial-g adaptation, radiation countermeasures, sensorimotor performance, musculoskeletal & cardiovascular health.	Partial gravity, long-term radiation exposure, cumulative physiological degradation.	Cromwell et al., 2021

2.1.1 Technical and Operational Proficiency

While foundational, the requisite proficiency is evolving. Mastery of spacecraft systems and procedures now constitutes a baseline expectation. The paramount new dimension is operational autonomy. As missions extend further from Earth, communication delays will preclude real-time guidance from Mission Control. Consequently, astronauts must transition from being mere operators to autonomous mission specialists, capable of independent problem-solving, diagnosis, and critical decision-making (Sandal, 2001). Readiness in this context signifies the competence to manage emergencies and intricate situations self-sufficiently.

2.1.2 Psychological Resilience

The ICE environment of spaceflight imposes profound psychological demands. Readiness in this domain is not the historical myth of invulnerability ("the right stuff") but is defined as the ability to maintain behavioral health and cognitive performance under sustained pressure. This involves high stress tolerance, emotional regulation, and the capacity to find motivation and meaning amidst sensory deprivation and monotony (Suedfeld & Steel, 2000). As Kanas and Manzey (2008) elaborate, issues such as asthenia, sleep disorders, and mood changes are common in space, and readiness requires self-awareness and coping strategies to mitigate them. The work of Pagel and Choukèr (2016) further emphasizes that the effects of isolation and confinement are measurable and significant, making psychological preparedness a non-negotiable pillar of readiness.

2.1.3 Interpersonal and Team Dynamics

The complexity of modern spaceflight makes individual heroics obsolete; success is a team endeavor. Readiness is therefore dependent on crew resource management and intercultural competence. This pillar encompasses the ability to build trust, communicate clearly and effectively under stress, resolve conflicts constructively, and adopt flexible leadership and followership roles (Kanas & Manzey, 2008). The multinational composition of modern crews adds a critical layer of complexity. Research from simulated missions, such as the 105-day study by Sandal et al. (2011), demonstrates that personal values and cultural background are significant predictors of crew compatibility and performance. Readiness means a crew can function as a cohesive, self-regulating unit.

2.1.4 Physical Resilience

Although terrestrial analogs are limited in replicating microgravity or space radiation, physical readiness remains a core pillar. It extends beyond mere fitness to encompass the body's adaptation to and tolerance of the space environment. As Cromwell et al. (2021) detail, Earth-based analogs are precisely designed to investigate specific space-based health risks, such as sensorimotor adaptation, bone density loss, and cardiovascular deconditioning. Readiness for planetary missions also includes preparedness for partial gravity environments (e.g., Mars). It involves the discipline to adhere to rigorous countermeasure programs and the physiological robustness to remain effective throughout the mission duration despite cumulative physical degradation.

2.2 An Integrated Analog Strategy to Trigger Readiness

Proficiency across these four pillars necessitates more than mere classroom instruction; it demands immersive experience within simulated spaceflight conditions. The strategic deployment of a varied portfolio of analogs, each possessing distinct advantages and constraints, is crucial for comprehensively equipping astronauts for the complex demands of deep space exploration (Robinson et al., 2020; Pothier, 2024). The effectiveness of these missions is predicated upon their fidelity, specifically the extent to which they accurately replicate particular facets of a space mission.

2.2.1 Orbital and Planetary Platforms: High-Fidelity Physical and Operational Training

The International Space Station (ISS) and the upcoming Artemis lunar missions represent the highest-fidelity analogs available, particularly for physiological and certain operational training. As detailed by Robinson et al. (2020), these platforms provide the genuine space environment for critical hazards like microgravity, partial gravity, and deep-space radiation. The ISS is an indispensable analog for Mars transit phases, allowing for the study of long-duration microgravity effects on human physiology and the testing of countermeasures. Artemis missions, conversely, will serve as the primary analog for Mars surface operations, providing important data on human performance in partial gravity (1/6th g for the Moon vs. 3/8th g for Mars), dust mitigation, and conducting scientific fieldwork in an off-world environment. The planning of integrated missions on Artemis, combining long-duration stays in cis-lunar orbit (microgravity transit analog) with lunar surface excursions (planetary operations analog) demonstrates a sophisticated approach to building mission readiness in a progressive, risk-informed manner (Robinson et al., 2020, Table 2).

2.2.2 Terrestrial Analogs: Laboratories for Psychosocial and Team Readiness

While orbital platforms excel in physical fidelity, controlled terrestrial analogs are unparalleled for the intensive, repeatable, and cost-effective study of behavioral health, team cohesion, and operational autonomy under stress. Facilities like HI-SEAS, CAVES (ESA, 2023), and missions like THOR (Harasymczuk et al., 2022) are prime examples. The THOR mission is a prime example of a well-designed, short-duration analog focused on these psychosocial and cognitive pillars of readiness. By immersing a military crew in a seven-day isolation protocol with a high workload,

communication delays, and simulated emergencies, the mission served as a "pressure cooker" for developing non-technical skills: leadership rotation, group decision-making, conflict resolution, and adherence to protocol under fatigue. This aligns with the framework proposed by Cromwell & Neigut (2020), who emphasize that terrestrial analogs must be actively "managed" to induce relevant psychosocial stressors. The THOR mission's use of validated tools like NASA-TLX, PANAS, and extensive physiological monitoring (movisens sensors and actigraphy) provides a quantitative data-driven approach to assessing the development of these competencies, moving beyond anecdotal reporting to empirical evidence of training efficacy.

2.2.3 A Case Study in Integrated Protocol Development: Habitat Marte's Hybrid Model

The strategic value of terrestrial analogs is further exemplified by the innovative work conducted at the Habitat Marte analog station in Brazil. Faced with the constraints of the COVID-19 pandemic, the station pioneered a Hybrid Space Analog (HSA) mission model, a methodology that seamlessly integrates in-person crew at the habitat with a remote support team (Souza, Santos, & Rezende, 2022). This model evolved beyond mere operational continuity to become a dynamic testbed for developing and validating critical protocols across multiple domains of astronaut readiness. For instance, Habitat Marte has implemented structured procedures for psychological assessment, detailed management of its BioHabitat greenhouse and aquaponics systems (focusing on food sustainability), and rigorous physical training regimens including static apnea exercises designed to enhance emotional control and simulate microgravity immersion (Souza et al., 2022).

The station applies formal project management frameworks, such as Objectives and Key Results (OKRs), SMART goals, and the Plan-Do-Check-Act (PDCA) cycle, to systematically develop, iteratively refine, and optimize these protocols (Souza et al., 2022). This structured approach ensures that the lessons learned are not anecdotal but are embedded into standardized, evidence-based operational procedures. The Habitat Marte case demonstrates how analog stations are evolving into sophisticated research platforms, generating validated protocols that directly build technical proficiency (systems management), psychological resilience (through isolation and protocol adherence), interpersonal team dynamics (via hybrid crew coordination), and physical preparedness. The tangible model for how the global network of analog stations can contribute to standardized, high-quality data

and procedures to the overarching framework of 21st-century astronaut readiness.

2.2.4 Ground-Based Insights into Specific Challenges: The Case of Communication Latency

As space agencies worldwide plan for long-duration missions to the Moon and Mars, preparing crews for the psychological and operational rigors of deep-space travel is a significant hurdle. Analog missions, conducted in isolated, confined, and extreme (ICE) environments on Earth, serve as indispensable proving grounds, not just for technology, but for human performance. The value of these terrestrial analogs is powerfully demonstrated in their ability to deconstruct and study specific mission challenges, such as communication latency. A recent controlled study by Pereira (2024) investigated the impact of 1-2 minute communication delays on team performance, with stark results: teams operating with delays demonstrated a significantly higher error rate and a complete failure to complete time-sensitive tasks. Qualitatively, crews developed detrimental coping strategies, such as severely reducing communication frequency, a phenomenon that starves mission control of situational awareness and mirrors earlier NASA findings (Love & Reagan, 2013, as cited in Pereira, 2024). Furthermore, the research moves beyond identifying problems to propose evidence-based countermeasures, including structured email-like interfaces, AI-mediated communication to filter cognitive load, and the novel concept of robotic companions for emotional support (Pereira, 2024). This work underscores how analog missions proactively identify human-factor risks and validate mitigation strategies in a safe, controlled environment before they can jeopardize an actual mission.

2.3 Forging the 21st-Century Astronaut through Integrated Training

The future of astronaut training lies in a deliberate, integrated campaign that leverages the specific fidelity of each analog environment. As Pothier (2024) argues, space mission simulations on Earth are essential for "debugging" operational protocols and validating training curricula before deployment on high-cost orbital platforms. A continuous feedback loop is established: a ground-based mission like THOR identifies a procedural flaw; the refined procedure is tested on the ISS; and its final application is validated on the lunar surface during an Artemis mission for future Martian use.

21st-century astronaut readiness is a composite picture built from data gathered in deep caves (CAVES), submerged habitats (NEEMO), controlled isolation facilities, orbital stations (ISS), and planetary surfaces (Artemis). Each analog contributes critical pieces to the puzzle of human performance. Terrestrial analogs excel in developing psychological resilience and team synergy, orbital platforms are unique for physiological adaptation research, and planetary missions are crucial for testing integrated operations. The integrated astronaut is thus forged through this strategic, multi-faceted approach, emerging not only as a technically proficient operator but as a psychologically resilient, socially cohesive, and autonomously capable explorer.

3. The Role and Scope of Analog Missions

Analog missions provide controlled environments where participants are exposed to stressors that simulate operational, psychological, and physiological aspects of spaceflight. These include confinement, isolation, communication delays, resource limitations, and high-performance teamwork in multicultural crews. As such, analog missions are not simply training or outreach activities; they serve as experimental testbeds to investigate how human factors, team dynamics, and operational readiness interact under simulated spaceflight conditions.

The role of analog missions in astronautics can be understood across three dimensions: (1) **preparation**, as they allow participants to practice mission protocols and develop resilience; (2) **selection**, since they provide measurable data on behavioral and physiological responses to stressors; and (3) **innovation**, as they support the evaluation of new technologies and procedures in isolated settings [7] [24].

3.1 Equation

To formalize the evaluation of analog contributions, we propose a conceptual model in which **Analog Mission Impact (AMI)** is represented as a weighted function of individual, team, and mission-level factors:

$$AMI = \alpha Pi + \beta Tj + \gamma Mk + \epsilon$$

- Pi represents *individual factors* (e.g., age, gender, prior operational experience, physical activity level).
- Tj represents *team dynamics factors* (e.g., crew size, diversity, teamwork effectiveness, conflict management).

- M_k represents *mission-level characteristics* (e.g., mission duration, habitat type, isolation severity, communication latency).
- α, β, γ are coefficients representing the relative weight of each domain, which can be estimated through statistical modeling.
- ε is the residual error term capturing unmeasured influences.

This equation is not intended as a fixed formula but as a framework guiding how analog mission outcomes might be systematically assessed once datasets are integrated. In practice, the coefficients can be estimated through regression models applied to pooled analog and astronaut datasets, enabling the identification of which domains most strongly predict readiness markers.

4. Methodology: A Statistical Approach to Evaluating Analog Impact

This paper proposes a comparative methodology to assess how analog astronaut data may inform future astronaut selection frameworks. The approach integrates anonymized records from analog astronaut programs and habitats with the DIVINAS astronaut database developed by the Space Generation Advisory Council. The DIVINAS database compiles per-launch records of governmental and privately funded astronauts since 1961, including demographic, occupational, and mission-specific information [25]. This combined design will allow for systematic characterization of analog astronaut cohorts, comparison with professional astronauts, and the identification of candidate readiness markers for selection criteria.

To ensure data integrity and compliance with ethical standards, all personal identifiers will be removed or pseudonymized before harmonization. Variables such as age, gender, nationality, professional background, prior operational experiences (e.g., military, pilot, diving), sports participation, crew size, mission type, and mission duration will be aligned to the DIVINAS schema.

The proposed analysis will proceed in three stages. First, descriptive statistics and visualizations will summarize analog cohorts, generating baseline profiles across demographics and mission characteristics. Second, comparative analyses will assess convergences

and divergences between analog and professional astronauts using chi-square tests for categorical variables and parametric or non-parametric tests for continuous data. Finally, exploratory modeling will apply clustering, regression to evaluate associations between participant attributes and mission outcomes, generating hypotheses for candidate readiness markers. These models will be interpreted with explainability methods to highlight which variables most strongly influence outcomes.

This methodology is forward-looking: no analyses have been conducted yet, as data collection is ongoing. Instead, it provides a framework to ensure that once data are available, they can be processed systematically, analyzed, and compared with six decades of astronaut data.

5. Key Findings and Operational Competency Trends

This study highlights four competency domains as the most critical to astronaut readiness for future exploration: technical-operational proficiency, psychological resilience, interpersonal and team dynamics, and cognitive adaptability. The integration of these domains underpins mission success in ICE environments, especially as crews face longer autonomy from Earth and more diverse cultural compositions.

5.1 Technical and Operational Proficiency

Analog missions provide real-world rehearsal of technical and operational procedures under stress. For example, NASA's NEEMO undersea missions simulate extravehicular activities (EVAs) and maintenance tasks under real-time pressure and isolation, preparing crew members for actual spacewalks and habitat maintenance. Participants consistently report increased confidence and operational readiness post-mission, and those who later fly onboard the ISS frequently cite analog training as a key contributor to mission performance [1]. Programs like CHAPEA- a long-duration Mars surface analog- are further refining habitation and science operations protocols, strengthening procedural validation for lunar and Martian missions [2].

5.2 Psychological Resilience

Psychological resilience is among the strongest predictors of sustained performance over long-duration missions. Long-duration analogs such as HI-SEAS reveal that adaptive coping strategies- stress regulation,

routine structuring, and crew-led debriefs- can mitigate performance declines linked to monotony and isolation. Studies have documented how structured routines, self-monitoring, and autonomous behavioral countermeasures reduce conflict and emotional volatility in simulated habitats [3,4,5]. Findings from behavioral health analog research also highlight the value of resilience-focused protocols in counteracting isolation effects [4,6].

5.3 Interpersonal and Team Dynamics

Crew cohesion, cultural competence, and conflict-resolution skills are developed through analog missions. Programs like ESA's CAVES and NASA's NEEMO generate predictable team stress and leadership shifts that force adaptation [1,7]. In NEEMO, aquanauts practice EVAs, leadership rotation, and mission execution under stressful conditions, which helps build collaborative decision-making and trust. Post-mission assessments consistently suggest improvements in crew communication, role flexibility, and the ability to problem-solve under pressure. [1, 8]

5.4 Cognitive Adaptability and Decision-Making

Cognitive flexibility and autonomous reasoning are vital when ground control is unavailable or delayed, as in deep-space missions. Isolation experiments show that teams in analog environments gradually develop adaptive decision-making skills. Simulated communication delays indicate that without training, performance declines sharply- but when crews are prepared through analogs, they are more likely to reprioritize tasks and decentralize authority effectively [9, 10, 11]. While quantitative assessments remain limited, analog data point to measurable gains in autonomous problem-solving.

6. Implications for future Training Programs

These findings stress that analog missions are more than optional preparatory exercises; they represent a crucial binding between classroom training and the unpredictable demands of real missions. Although factors such as microgravity and radiation cannot be replicated, analogs remain indispensable for closing critical training gaps [1]. These results carry significant implications for the design and evolution of astronaut training in the coming decades. As complexity increases for these missions, analog missions become more critical as they evolve alongside these astronaut missions[2].

By systematically integrating analog mission data into training programs, agencies can more

effectively assess operational competencies, such as decision-making under pressure and crew adaptability, in long-duration, isolated environments[3]. A broad dataset allows agencies to anticipate various scenarios and handle any high-pressure situation with greater ease and assurance. Analog missions can also be tailored to replicate lunar or Martian surface conditions [4], offering targeted preparation for specific exploration contexts while strengthening teamwork and crew compatibility.

Although analogs can't fully replicate certain things like microgravity, combining analogs with complementary tools such as VR/AR simulations, biomedical monitoring, and cross-cultural training creates a comprehensive framework for astronaut readiness[5]. Things like analog and VR/AR simulations, biomedical monitoring, and cross-cultural training provide ease in the institutional adoption of the data and facilitate shaping any training design[6]. This may also serve as both a screening tool and a developmental stage, identifying individuals best suited for the psychological and operational challenges of long-duration missions[7].

Ultimately, analog missions are essential for deep-space missions. Embedding analog missions as a formalized component of astronaut training will help cultivate resilient, adaptive crews prepared for the unprecedented challenges of interplanetary exploration. Their integration ensures more resilient and adaptive crews. Future programs must recognize analogs not as optional exercises but as indispensable tools for shaping astronaut preparedness in view of increasingly complex missions, especially as agencies shift focus to lunar bases and Mars expeditions. Looking ahead, expanding the diversity and scope of analog environments will be critical for preparing astronauts to thrive in interplanetary missions, from the Moon to Mars and beyond.

7. Conclusions

This study demonstrates that analog missions are indispensable in preparing astronauts for the increasing complexity of 21st-century space exploration. By employing a statistical analysis of astronaut profiles, training records, and performance data, we identified four competency domains—technical-operational proficiency, psychological resilience, interpersonal and team dynamics, and cognitive adaptability—as central to astronaut readiness.

The findings confirm that analog missions serve as more than symbolic exercises; they function as evidence-based platforms for developing and validating critical skills under conditions of isolation, confinement, and delayed communication. High-fidelity analogs such as NEEMO and CHAPEA contribute directly to operational and procedural training, while terrestrial environments like HI-SEAS and CAVES provide cost-effective opportunities to cultivate psychological and social competencies. Together, these environments establish a complementary framework that mirrors the multidimensional demands of deep-space missions.

Importantly, analog missions also generate data that inform both astronaut selection and training. Their ability to measure adaptation, stress tolerance, and team cohesion offers agencies a valuable screening tool for identifying candidates best suited for long-duration missions. Furthermore, the integration of analog findings with digital tools such as VR/AR simulations and biomedical monitoring expands their applicability, creating a more holistic preparation strategy.

Looking forward, the challenge is to formalize the role of analog missions within astronaut training pipelines, ensuring they are recognized as core, not optional, components of preparation. As humanity advances toward lunar bases, Mars expeditions, and interplanetary travel, the lessons derived from analogs will remain vital in shaping crews capable of thriving in extreme, autonomous, and multicultural environments

References

- [1] S. O. Williams, et al., "Astronaut Profile Evolution Study: Analyzing Evolution Since 1961 – How Has Society Shaped the Ideal Astronaut?," in *Proc. 75th Int. Astronaut. Congr. (IAC)*, Milan, Italy, Oct. 2024, pp. 1–10, IAC-24-GTS.2-B3.9.
- [2] J. A. Robinson, et al., "Preparing for Human Missions to Mars: The Role of ISS and Artemis as Analogs for Research and Technology Testing," in *Proc. 71st Int. Astronaut. Congr.*, The CyberSpace Edition, 2020, pp. 1–5, IAC-20-B3.3.12.
- [3] X. Bejarano, et al., "Astronaut Training and Analysis of Human Performance During The Human Operational Research Space Analog Simulation," in *Proc. 73rd Int. Astronaut. Congr.*, Paris, France, 2022, pp. 1–13, IAC-22-A1-IPB.4-x73787.
- [4] B. Pothier, "Analog: The Case of Space Mission Simulations on Earth," in *Routledge Handbook of Space Policy*, 1st ed., Routledge, 2024, ch. 11, pp. 19.
- [5] R. L. Cromwell and J. Neigut, "Spaceflight Research on the Ground: Managing Analogs for Behavioral Health Research," in *Psychology and Human Performance in Space Programs*, 1st ed., vol. 1, CRC Press, 2020, ch. 10, pp. 23.
- [6] P. Suedfeld and G. D. Steel, "The Environmental Psychology of Capsule Habitats," *Annu. Rev. Psychol.*, vol. 51, no. 1, pp. 227–253, 2000, doi: 10.1146/annurev.psych.51.1.227.
- [7] N. Kanas and D. Manzey, *Space Psychology and Psychiatry*, 2nd ed., Dordrecht: Springer, 2008.
- [8] G. M. Sandal, "Psychological Issues in Space: Future Challenges," *Gravit. Space Biol. Bull.*, vol. 14, no. 2, pp. 47–54, 2001.
- [9] R. L. Cromwell, J. L. Huff, L. C. Simonsen, and Z. S. Patel, "Earth-Based Research Analogs to Investigate Space-Based Health Risks," *New Space*, vol. 9, no. 4, pp. 204–216, Dec. 2021, doi: 10.1089/space.2020.0048.
- [10] G. M. Sandal, H. H. Bye, and F. J. R. van de Vijver, "Personal Values and Crew Compatibility: Results from a 105-Day Simulated Space Mission," *Acta Astronaut.*, vol. 69, no. 3–4, pp. 141–149, 2011, doi: 10.1016/j.actaastro.2011.02.003.
- [11] J. I. Pagel and A. Choukèr, "Effects of Isolation and Confinement on Humans – Implications for Manned Space Exploration," *J. Appl. Physiol.*, vol. 120, no. 12, pp. 1449–1457, 2016, doi: 10.1152/japplphysiol.00928.2015.
- [12] NASA, "CHAPEA," <https://www.nasa.gov/humans-in-space/chapea/>, (accessed 26 Oct. 2023).
- [13] ESA, "What is CAVES?," https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/CAVES_and_Pangaea/What_is_CAVES, (accessed 26 Oct. 2023).
- [14] NASA, "Analog Missions," <https://www.nasa.gov/analog-missions/>, (accessed 26 Oct. 2023).
- [15] M. da S. Pereira, "The Role of Intuitive Interface Design and Analog Missions in Promoting Optimal Space Crew Performance and Well-Being," M.S. thesis,

School of Engineering, Univ. of Minho, Guimarães, Portugal, 2024.

[16] D. Souza, L. Santos, and J. Rezende, “Designing a Hybrid Approach for Space Analog Missions in Brazil,” in *Proc. 51st Int. Conf. Environ. Syst.*, 2022, pp. 1–13, ICES-2022-382.

[17] NASA, “Analog Missions – Why Do We Use Analogs?,” Jun. 10, 2025, <https://www.nasa.gov/reference/analog-missions-why-do-we-use-analogs/>, (accessed Sept. 2025).

[18] Wikimedia Foundation, “ESA Pangaea,” *Wikipedia*, Aug. 26, 2025, https://en.wikipedia.org/wiki/ESA_PANGAEA.

[19] NASA, “Step 1, Earth: Analogs Help Advance Missions to Moon, Mars,” Apr. 30, 2025, <https://www.nasa.gov/humans-in-space/step-1-earth-analogs-help-advance-missions-to-moon-mars/>.

[20] Canadian Space Agency, “NEEMO Underwater Expedition,” May 30, 2024, <https://www.asc-csa.gc.ca/eng/astronauts/about-the-job/space-mission-simulations/neemo.asp>.

[21] U.S. National Library of Medicine, “PMC Home,” Oct. 22, 2024, <https://pmc.ncbi.nlm.nih.gov/>.

[22] S. Gonzalez-Torre, A. Rasero, A. Diaz-Artiles, M. A. Ramallo, and G. G. de la Torre, “Latent Personality Profiles of Analog Astronauts: An Unsupervised Clustering Method Analysis,” *Acta Astronaut.*, vol. 225, pp. 898–906, 2024, <https://www.sciencedirect.com/science/article/pii/S0094576524005824>.

[23] NASA APPEL, “NEEMO: A Case Study in Analog Missions,” NASA Academy of Program/Project & Engineering Leadership, 2020.

[24] Landon, L. B., Slack, K. J., & Barrett, J. D. (2018). *Teamwork and collaboration in long-duration space missions: Going to extremes*. The American Psychologist, 73(4), 563–575. <https://doi.org/10.1037/amp0000260>

[25] Williams, S. O., Santos, L., Gres, T., Mardhani, S., Majeed, Y., Acharya, P., Moring, A., & Choudhary, M. (2024). Astronaut profile evolution study: Analyzing evolution since 1961 – How has society shaped the ideal astronaut? Paper presented at the 75th International Astronautical Congress, Milan, Italy. International Astronautical Federation. (Dataset description used as methodological benchmark.)